MATH 231 – Linear Algebra I

The Matrix Unit, Part 8 ▪ Solving Linear Systems: Elimination

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About this document. This is Part 8 of the matrix unit, Sections 40–45. Section numbers run continuously through the unit (Part 2's opening table lists them all).

Assumed. Linear systems and $\mathbf{Ax} = \mathbf{b}$ (Part 1); triangular matrices (§4.4); column space, null space, rank and rank–nullity (§10–§12, §31–§32); the determinant and its laws D1–D10 (§38–§39).

What you should be able to do afterwards. Perform the three elementary row operations and write each as a matrix; solve triangular systems by back and forward substitution; reduce a matrix to row-echelon form by Gaussian elimination and solve the system; continue to reduced row-echelon form by Gauss–Jordan elimination; read off the rank, the pivot and free variables, a basis of the null space, and every solution; decide from the reduced form whether a system has no solution, one, or infinitely many; compute a determinant by elimination; and describe what to expect of over- and underdetermined systems.

Part 1 wrote a linear system as $\mathbf{Vx} = \mathbf{u}$, solved a 2×2 example by substitution, and noted that substitution is not a method that survives twenty equations. It promised a procedure that works on the matrix directly, applies to any size, and ends either with the answer or with a definite report that there is none. Since then, Parts 2–7 have explained what the possible answers are. This part supplies the procedure.

40. Elementary row operations

40.1. The augmented matrix

The system $\mathbf{Ax} = \mathbf{b}$ is completely described by the coefficients and the right-hand side. Writing $\mathbf{b}$ as an extra column, separated by a bar, gives the **augmented matrix** $[\,\mathbf{A} \mid \mathbf{b}\,]$. Each row is one equation:

$$\begin{aligned} x_1 + 2x_2 &= 5 \\ x_2 + x_3 &= 5 \\ x_1 + x_3 &= 4 \end{aligned} \qquad \longleftrightarrow \qquad \left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 1 & 1 & 5 \\ 1 & 0 & 1 & 4 \end{array}\right].$$

This is the system of Part 1 §1.6, whose solution was constructed to be $\mathbf{x} = [\,1, 2, 3\,]$. Solving it, by a method, is §42's first job.

40.2. Three operations

Definition 40.1 (Elementary row operations)**.** The three **elementary row operations** on a matrix are:

(E1) **swap** two rows;

(E2) **scale** a row by a nonzero number;

(E3) **replace** a row by itself plus a multiple of another row.

On the augmented matrix these are things one does to equations without thinking: list them in a different order, multiply one through by a nonzero number, add a multiple of one equation to another.

40.3. They do not change the solutions

Theorem 40.1. *If* $[\,\mathbf{A}' \mid \mathbf{b}'\,]$ *is obtained from* $[\,\mathbf{A} \mid \mathbf{b}\,]$ *by an elementary row operation, the systems* $\mathbf{Ax} = \mathbf{b}$ *and* $\mathbf{A}'\mathbf{x} = \mathbf{b}'$ *have exactly the same solutions.*

Proof. Every equation of the new system is a combination of equations of the old one, so any solution of the old system satisfies the new one. And every operation can be undone by an operation of the same kind: swap the same rows back; scale by the reciprocal; subtract the multiple that was added. So the old system is also obtained from the new one, and any solution of the new system satisfies the old one. □

40.4. Each operation is a matrix

Doing a row operation to $\mathbf{A}$ is the same as multiplying $\mathbf{A}$ on the left by the matrix $\mathbf{E}$ obtained by doing that operation to $\mathbf{I}$. Row i of $\mathbf{EA}$ is (row i of $\mathbf{E}$) times $\mathbf{A}$, a combination of the rows of $\mathbf{A}$ with coefficients from row i of $\mathbf{E}$. In $\mathbb{R}^{3\times 3}$:

$$\underbrace{\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}}_{\text{swap rows 1, 2}}, \qquad \underbrace{\begin{bmatrix}1&0&0\\0&c&0\\0&0&1\end{bmatrix}}_{\text{scale row 2 by } c}, \qquad \underbrace{\begin{bmatrix}1&0&0\\0&1&0\\k&0&1\end{bmatrix}}_{\text{add } k \text{ row 1 to row 3}}.$$

These are **elementary matrices**. They are old friends from Part 3: a swap is a reflection (it exchanges two coordinates), a scaling is a non-uniform scaling, and a replacement is a shear. Their determinants are -1, c and 1, which are the laws D5, D4 and D8 of §38. For example, with $k = -1$,

$$\begin{bmatrix}1&0&0\\0&1&0\\-1&0&1\end{bmatrix}\begin{bmatrix}1&2&0\\0&1&1\\1&0&1\end{bmatrix} = \begin{bmatrix}1&2&0\\0&1&1\\0&-2&1\end{bmatrix},$$

which is "subtract row 1 from row 3".

41. Triangular systems

41.1. Back substitution

A triangular system is the easy case, and the goal of elimination is to reach one. If $\mathbf{U}$ is upper triangular, the last equation of $\mathbf{Ux} = \mathbf{c}$ involves only x_n; the one above it, only x_{n-1} and x_n; and so on. Solve from the bottom up.

Example 41.1.

$$\left[\begin{array}{ccc|c}2&1&-1&3\\0&3&2&7\\0&0&4&8\end{array}\right] \qquad \begin{aligned} x_3 &= 8/4 = 2,\\ x_2 &= (7 - 2x_3)/3 = (7-4)/3 = 1,\\ x_1 &= (3 - x_2 + x_3)/2 = (3 - 1 + 2)/2 = 2. \end{aligned}$$

Check in the first equation: $2(2) + 1 - 2 = 3$. ✓

In general, once $x_{i+1}, \ldots, x_n$ are known, equation i gives

$$x_i = \frac{1}{u_{ii}}\Big(c_i - \sum_{j=i+1}^{n} u_{ij}x_j\Big).$$

Back substitution

Input: an upper triangular $\mathbf{U} \in \mathbb{R}^{n\times n}$ with every $u_{ii} \neq 0$, and $\mathbf{c} \in \mathbb{R}^n$. **Output:** the solution $\mathbf{x}$ of $\mathbf{Ux} = \mathbf{c}$.

for $i = n, n-1, \dots, 1$:
  $s \leftarrow c_i$
  for $j = i+1, \dots, n$:
    $s \leftarrow s - u_{ij}\, x_j$ *remove the unknowns already found*
  $x_i \leftarrow s \,/\, u_{ii}$
return $\mathbf{x}$

41.2. Forward substitution

For a lower triangular $\mathbf{L}$, the first equation involves only x_1, so solve from the top down: $x_i = \frac{1}{l_{ii}}\big(c_i - \sum_{j<i} l_{ij}x_j\big)$ for $i = 1, \dots, n$.

Example 41.2.

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 2 & 1 & 0 & 4 \\ -1 & 3 & 2 & 7 \end{array}\right] \qquad x_1 = 1, \quad x_2 = 4 - 2(1) = 2, \quad x_3 = \big(7 + 1 - 3(2)\big)/2 = 1.$$

41.3. When it works, and what it costs

Substitution divides by every diagonal entry, so it needs them all nonzero. For a triangular matrix that is exactly the condition $\det \mathbf{U} = u_{11} \cdots u_{nn} \neq 0$ (§39.1). If some $u_{ii} = 0$, the system has either no solution or infinitely many, and §43 shows how to tell which.

Row i costs about $n - i$ multiplications and as many subtractions, so the whole substitution costs about $1 + 2 + \cdots + n \approx \frac{n^2}{2}$ of each, roughly n^2 operations. For $n = 1000$ that is a million operations, a few milliseconds of computing time. Triangular systems are cheap.

42. Gaussian elimination and row-echelon form

42.1. The idea

Use row operations, which do not change the solutions, to turn $[\,\mathbf{A} \mid \mathbf{b}\,]$ into $[\,\mathbf{U} \mid \mathbf{c}\,]$ with $\mathbf{U}$ upper triangular, then back substitute. Part 1 §4.4 hinted at this when it introduced triangular matrices. The method works one column at a time: use the diagonal entry of the column, the **pivot**, to clear every entry below it to zero.

42.2. Part 1's system, solved

Example 42.1. Start from the augmented matrix of §40.1.

Column 1. The pivot is the 1 in row 1. Row 2 already has a 0 there. Row 3 has a 1, so subtract row 1 from row 3:

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 1 & 1 & 5 \\ 1 & 0 & 1 & 4 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 1 & 1 & 5 \\ 0 & -2 & 1 & -1 \end{array}\right].$$

Column 2. The pivot is the 1 in row 2. Clear the -2 below it by adding twice row 2 to row 3:

$$\xrightarrow{R_3 \leftarrow R_3 + 2R_2} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 3 & 9 \end{array}\right].$$

Upper triangular. Back substitute: $x_3 = 9/3 = 3$, $x_2 = 5 - x_3 = 2$, $x_1 = 5 - 2x_2 = 1$. So $\mathbf{x} = [\,1, 2, 3\,]$, the coordinates Part 1 §1.6 built the example from, now *found* rather than known in advance.

42.3. The algorithm

For an $n \times n$ system, the multiplier that clears entry (i, k) is $m_{ik} = a_{ik}/a_{kk}$, the entry divided by the pivot above it.

Gaussian elimination (forward phase)

Input: the augmented matrix $[\,\mathbf{A} \mid \mathbf{b}\,]$ with $\mathbf{A} \in \mathbb{R}^{n \times n}$. **Output:** $[\,\mathbf{U} \mid \mathbf{c}\,]$ with $\mathbf{U}$ upper triangular and the same solutions.

1: **for** $k = 1, \ldots, n-1$: *work on column k*
2: **if** $a_{kk} = 0$:
3: **if** a lower row has a nonzero entry in column k: swap it with row k
4: **else**: **continue** *no pivot in this column*
5: **for** $i = k+1, \ldots, n$:
6: $m \leftarrow a_{ik} \,/\, a_{kk}$
7: row $i \leftarrow$ row $i - m \cdot$ row k *now $a_{ik} = 0$*
8: **return** $[\,\mathbf{U} \mid \mathbf{c}\,]$

Then back substitution finishes the job, if every diagonal entry of $\mathbf{U}$ is nonzero. Counting operations: clearing column k updates $n-k$ rows of about $n-k$ entries each, about $2(n-k)^2$ operations, and summing over k gives about $\frac{2}{3}n^3$. Elimination is the expensive part. For $n = 1000$ it is under a billion operations, which a laptop does in about a second.

42.4. Zero pivots, and small ones

Example 42.2. $\left[\begin{array}{cc|c} 0 & 1 & 2 \\ 1 & 1 & 3 \end{array}\right]$: the first pivot position holds a 0, and nothing can be divided by it. Line 3 swaps the rows: $\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 2 \end{array}\right]$, already triangular, so $x_2 = 2$ and $x_1 = 1$.

Remark. On a computer, a pivot that is merely *small* is nearly as bad as a zero one: dividing by it magnifies every rounding error in its row. So practical codes swap at every column, bringing up whichever entry at or below the diagonal is largest in absolute value. This is *partial pivoting*, and it is how the algorithm is actually run.

42.5. Row-echelon form

Elimination works on any matrix, square or not, singular or not, if it is allowed to move right whenever a column has no pivot. What it produces has a definite shape.

Definition 42.1 (Row-echelon form)**.** A matrix is in **row-echelon form** if

(1) any rows of all zeros are at the bottom, and

(2) the first nonzero entry of each nonzero row (its **pivot**) lies strictly to the right of the pivot of the row above.

The columns that contain pivots are the **pivot columns**.

The name is a picture: the pivots descend in steps, like an echelon of troops. ($*$ marks any number; boxes mark pivots.)

$$\begin{bmatrix} \boxed{2} & * & * & * \\ 0 & \boxed{1} & * & * \\ 0 & 0 & 0 & \boxed{5} \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ is in row-echelon form;} \qquad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ are not.}$$

For a square matrix, row-echelon form is upper triangular.

42.6. The determinant, by elimination

For a square $\mathbf{A}$, elimination uses only swaps (each changes the sign of the determinant, D5) and replacements (no change, D8), and ends at an upper triangular $\mathbf{U}$ whose determinant is the product of its diagonal (§39.1). So

$$\boxed{\det \mathbf{A} = (-1)^s \, u_{11} u_{22} \cdots u_{nn},}$$

where s is the number of swaps. In Example 42.1 there were no swaps and the diagonal was $1, 1, 3$, so $\det \mathbf{A} = 3$. A cofactor expansion agrees: $1(1-0) - 2(0-1) + 0 = 3$. At about $\frac{2}{3}n^3$ operations, this is how determinants are computed in practice, not by the $n!$ of cofactors.

This also completes law D10 of §38. Row operations do not change solutions, so $\text{null}[\mathbf{A}] = \text{null}[\mathbf{U}]$. If every $u_{ii} \neq 0$, back substitution on $\mathbf{Ux} = \mathbf{0}$ gives only $\mathbf{x} = \mathbf{0}$. If some $u_{kk} = 0$ (take the first), the first k columns of $\mathbf{U}$ have nonzero entries only in the first $k-1$ rows, and k vectors confined to $k-1$ coordinates are dependent. So

$$\text{null}[\mathbf{A}] = \{\mathbf{0}\} \iff \text{every } u_{ii} \neq 0 \iff \det \mathbf{A} \neq 0,$$

and with rank–nullity, $\text{rank}\, \mathbf{A} = n \iff \det \mathbf{A} \neq 0$.

43. Gauss–Jordan elimination and reduced row-echelon form

43.1. Going further

Gaussian elimination stops at triangular and hands over to back substitution. **Gauss–Jordan elimination** keeps going with row operations instead: it scales every pivot to 1 and clears the entries *above* each pivot as well as below.

Definition 43.1 (Reduced row-echelon form)**.** A matrix is in **reduced row-echelon form** (RREF) if it is in row-echelon form, every pivot equals 1, and every pivot is the only nonzero entry in its column.

Every matrix can be brought to reduced row-echelon form, and (we state without proof) the result does not depend on which row operations were used: a matrix has exactly one RREF.

43.2. A square system

Example 43.1. Continue Example 42.1 from its triangular form:

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 3 & 9 \end{array}\right] \xrightarrow{R_3 \leftarrow \frac{1}{3} R_3} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$$\xrightarrow{R_2 \leftarrow R_2 - R_3} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right] \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

The left block is now $\mathbf{I}$, and the system reads $x_1 = 1$, $x_2 = 2$, $x_3 = 3$ directly.

For a square system with a unique solution, Gauss–Jordan ends at $[\,\mathbf{I} \mid \mathbf{x}\,]$. Its real advantage shows when the solution is *not* unique.

43.3. Pivot variables and free variables

Example 43.2. Three equations, four unknowns:

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 1 \\ 2 & 4 & 1 & 4 & 4 \\ 3 & 6 & 1 & 5 & 5 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3 - 3R_1]{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 & 2 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - R_2} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Column 2 had no pivot (after column 1 was cleared, its entries below row 1 were all 0), so the second pivot is in column 3. The result is already in reduced row-echelon form, and it says

$$x_1 + 2x_2 + x_4 = 1, \qquad x_3 + 2x_4 = 2, \qquad 0 = 0.$$

The pivot columns are 1 and 3; their unknowns x_1, x_3 are **pivot variables**. The others, x_2 and x_4, are **free variables**: nothing constrains them. Set $x_2 = s$ and $x_4 = t$, and solve for the pivot variables: $x_1 = 1 - 2s - t$, $x_3 = 2 - 2t$. Every solution is

$$\mathbf{x} = \underbrace{\begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix}}_{\mathbf{x}_p} + s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ -2 \\ 1 \end{bmatrix}.$$

This is the shape promised in §31.6: one solution $\mathbf{x}_p$ (take $s = t = 0$) plus the null space. Check: $\mathbf{A}\mathbf{x}_p = [\,1,\ 2+2,\ 3+2\,] = [\,1, 4, 5\,]$, and each of the two other vectors is sent to $\mathbf{0}$.

43.4. Reading off rank, nullity and bases

Row operations do not change which combinations of the columns are zero, because $\mathbf{A}\mathbf{x} = \mathbf{0}$ and $\mathbf{R}\mathbf{x} = \mathbf{0}$ have the same solutions when $\mathbf{R}$ is the RREF of $\mathbf{A}$. In $\mathbf{R}$, the pivot columns are distinct standard basis vectors, so they are independent, and each non-pivot column is a combination of the pivot columns to its left (its entries are the coefficients). The same relations hold among the columns of $\mathbf{A}$. Hence:

Theorem 43.1. *For any matrix* $\mathbf{A}$*, with* $\mathbf{R}$ *its reduced row-echelon form:*

(a) *the columns of* $\mathbf{A}$ *in the pivot positions form a basis of* $\operatorname{col}[\mathbf{A}]$*, so* $\operatorname{rank}\mathbf{A}$ *is the number of pivots;*

(b) *setting one free variable to* 1 *and the others to* 0*, for each free variable in turn, gives a basis of* $\operatorname{null}[\mathbf{A}]$*, so* $\operatorname{nullity}\mathbf{A}$ *is the number of free variables.*

For (b), the vectors are independent because each has a 1 in its own free position, where all the others have 0.

Every column is either a pivot column or a free column, so

$$\underbrace{\#\text{pivots}}_{\text{rank}} + \underbrace{\#\text{free}}_{\text{nullity}} = n.$$

This is the rank–nullity theorem of §32 again, now proved by counting columns. In Example 43.2, rank = 2 (a basis of the column space is $[\,1,2,3\,]$ and $[\,0,1,1\,]$, columns 1 and 3 of $\mathbf{A}$), nullity = 2, and $n = 4$.

44. Homogeneous systems

44.1. Always consistent

A homogeneous system $\mathbf{Ax} = \mathbf{0}$ always has the **trivial solution** $\mathbf{x} = \mathbf{0}$. Its augmented column is zero and stays zero under every row operation, so in practice one just reduces $\mathbf{A}$. The only question is whether there are **nontrivial** solutions as well, and by Theorem 43.1 the answer is a count:

$$\text{nontrivial solutions exist} \iff \text{there is a free variable} \iff \operatorname{nullity}\mathbf{A} > 0 \iff \operatorname{rank}\mathbf{A} < n.$$

Theorem 44.1. *A homogeneous system with more unknowns than equations ($m < n$) always has nontrivial solutions.*

Proof. There is at most one pivot per row, so at most $m < n$ pivots, and at least one column is free. □

44.2. An example

Example 44.1. $x + y + z = 0$ and $x - y = 0$:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \end{bmatrix} \xrightarrow{R_2 \leftarrow -\frac{1}{2}R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & \frac{1}{2} \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \end{bmatrix}.$$

z is free; $x = -\frac{1}{2}z$ and $y = -\frac{1}{2}z$. With $z = 2t$, the solutions are $t[\,-1,-1,2\,]$, a line in $\mathbb{R}^3$. Check: $-1 - 1 + 2 = 0$ and $-1 + 1 = 0$.

44.3. When a system is consistent

For a general $\mathbf{b}$, elimination on $[\,\mathbf{A} \mid \mathbf{b}\,]$ settles everything. A system is **consistent** if it has at least one solution.

- If the row-echelon form has a row $[\,0 \;\cdots\; 0 \mid c\,]$ with $c \neq 0$, that row says $0 = c$, and there is **no solution**. Otherwise the system is consistent.
- Consistent with no free variable: **exactly one** solution.

- Consistent with a free variable: **infinitely many**.

A row $[\,0\ \cdots\ 0 \mid c\,]$ with $c \neq 0$ means the last column of $[\,\mathbf{A} \mid \mathbf{b}\,]$ is a pivot column. Equivalently, appending $\mathbf{b}$ raised the rank, or $\mathbf{b} \notin \operatorname{col}[\mathbf{A}]$:

$$\mathbf{Ax} = \mathbf{b} \text{ is consistent } \iff \operatorname{rank}[\,\mathbf{A} \mid \mathbf{b}\,] = \operatorname{rank}\mathbf{A} \iff \mathbf{b} \in \operatorname{col}[\mathbf{A}].$$

Example 44.2. §2.6 P5, one last time: $\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 4 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & -5 \end{array}\right]$. The second row says $0 = -5$: no solution, found by the algorithm without any picture.

45. Overdetermined and underdetermined systems

45.1. More equations than unknowns

A system with $m > n$ is **overdetermined**. Its column space has dimension at most $n < m$, a thin subspace of $\mathbb{R}^m$, and a randomly chosen $\mathbf{b}$ almost never lies in it. Overdetermined systems are usually inconsistent, and they are what measured data produce.

Example 45.1. Find a line $y = c_0 + c_1 t$ through the three data points $(1,1)$, $(2,2)$, $(3,2)$. Each point gives one equation in the two unknowns c_0, c_1:

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 3 & 2 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 1 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{array}\right].$$

The last row says $0 = -1$. No line passes through all three points, which is obvious once they are plotted. Yet there is clearly a line that passes *close* to all three. Finding it is the subject of Part 9, which uses these same three points.

45.2. Fewer equations than unknowns

A system with $m < n$ is **underdetermined**. By §44.1 the null space is nonzero, with nullity at least $n - m$, so if the system is consistent it has infinitely many solutions.

Example 45.2. $x_1 + x_2 + x_3 = 3$: one equation, three unknowns, a whole plane of solutions, $[\,3,0,0\,] + s[\,-1,1,0\,] + t[\,-1,0,1\,]$. Among them, $[\,3,0,0\,]$ has length 3 and $[\,1,1,1\,]$ has length $\sqrt{3}$. Is there a best one? A natural choice is the *shortest*, and §54 shows it is $[\,1,1,1\,]$, the solution lying along the row $[\,1,1,1\,]$ itself.

45.3. Summary

Shape	Typical behaviour	What decides it
square, rank $= n$	exactly one solution, for every $\mathbf{b}$	$\det \mathbf{A} \neq 0$
square, rank $< n$	none, or infinitely many	whether $\mathbf{b} \in \operatorname{col}[\mathbf{A}]$
overdetermined, $m > n$	usually none	whether $\mathbf{b} \in \operatorname{col}[\mathbf{A}]$; else least squares (Part 9)
underdetermined, $m < n$	none, or infinitely many	if consistent, choose the shortest (§54)

45.4. Practice

P1. Solve by Gaussian elimination and back substitution: $x + y + z = 6$, $2x + 3y + z = 11$, $x - y + 2z = 5$. Read off the determinant of the coefficient matrix from the pivots.

P2. Bring $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{array}\right]$ to reduced row-echelon form and write every solution as $\mathbf{x}_p$ plus the null space. Give the rank and nullity.

P3. Is $x + y = 2$, $x - y = 0$, $2x + y = 3$ consistent? If so, how many solutions?

P4. Write the elementary matrix that adds 5 times row 1 to row 3 of a 3×3 matrix, and the elementary matrix that undoes it.

P5. Compute $\det \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 1 \\ 2 & 0 & 3 \end{bmatrix}$ by elimination, and check by cofactors.

P6. Without computing, how many free variables at least does a homogeneous system of 3 equations in 5 unknowns have?

Answers

P1. $R_2 - 2R_1$: $[\,0, 1, -1 \mid -1\,]$; $R_3 - R_1$: $[\,0, -2, 1 \mid -1\,]$; $R_3 + 2R_2$: $[\,0, 0, -1 \mid -3\,]$. So $z = 3$, $y = -1 + 3 = 2$, $x = 6 - 2 - 3 = 1$. No swaps and pivots $1, 1, -1$, so $\det = -1$.

P2. $R_2 - 2R_1$: $[\,0, 0, 1 \mid 1\,]$; $R_1 - R_2$: $[\,1, 2, 0 \mid 2\,]$. RREF $\left[\begin{array}{ccc|c} 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right]$; $x_2 = s$ free, $x_1 = 2 - 2s$, $x_3 = 1$. $\mathbf{x} = [\,2, 0, 1\,] + s[\,-2, 1, 0\,]$. Rank 2, nullity 1.

P3. $R_2 - R_1$: $[\,0, -2 \mid -2\,]$; $R_3 - 2R_1$: $[\,0, -1 \mid -1\,]$; $R_3 - \frac{1}{2}R_2$: $[\,0, 0 \mid 0\,]$. Consistent, no free variable: exactly one solution, $y = 1$, $x = 1$. An overdetermined system *can* be consistent; this one happens to be.

P4. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 5 & 0 & 1 \end{bmatrix}$, undone by $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 0 & 1 \end{bmatrix}$.

P5. Swap R_1, R_2: $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$; $R_3 - 2R_1$: $[\,0, -2, 1\,]$; $R_3 + R_2$: $[\,0, 0, 2\,]$. Pivots $1, 2, 2$, one swap: $\det = -4$. Cofactors along row 1: $0 - 2(3 - 2) + 1(0 - 2) = -4$. ✓

P6. At most 3 pivots among 5 columns, so at least 2 free variables.

Looking ahead. Example 45.1 ended with no solution and an obvious best approximate one. Part 9 makes "best" precise: the $\mathbf{x}$ that makes $\mathbf{Ax}$ as close as possible to $\mathbf{b}$. It finds that $\mathbf{x}$ by solving a different, always-consistent square system, using the elimination of this part.